

Helvin Boateng

2026 FA [1] CIS 344 OS1C [51600] Database Desgn&Prog [Lecture] [Lehman College]

Professor Yanilda Peralta Ramos

09/29/26

Mini World: Music Band Website

1. Requirements Gathering

The data needed for the Music Band Website database was collected by looking for the key elements presented on several music bands' pages. A typical music band website contains information about its members, songs and albums, concerts, s and places. Therefore, representatives of these areas formed the basis of the database.

2. System Requirements

The database for the Music Band Website must be able to:

1. Hold information about each member of the group.
2. Note down data concerning the band's discs.
3. Retain records of songs.
4. Identify which song relates to which album.
5. Keep the details of the gigs taking place.
6. Log details about where the event takes place.
7. Track locations used for each concert.
8. Monitor members of the group who perform at each concert.

Band Member

- Member ID
- First Name
- Last Name
- Role
- Email
- Join Date

Album

- Album ID
- Album Title
- Release Date

Song

- Song ID
- Song Title
- Duration
- Album ID

Venue

- Venue ID
- Venue Name
- City
- State
- Capacity

Event

- Event ID
- Event Name
- Event Date
- Event Time
- Venue ID

Member Event

- Member ID
- Event ID
- Performance Role

Step 1 –Mini World Selection: I opted for the Music Band Website as my mini world due to its structured data that can easily be portrayed by means of a database.

Step 2 – Research: I investigated various types of information that music band websites have to organize. I found out that music band websites generally include information about the band members, the albums, the songs, the concerts, and all the venues.

Step 3 – Requirements: Using the research findings, I came to the conclusion that the system must store data on the band members, albums, songs, events, and venues. The system must also be able to track the music albums, venues of events, and band members performing at the events.

Step 4 – Entities: Based on requirements, I identified the key entities for the database. The entities include Band Member, Album, Song, Venue, Event, and Band Member Event.

Step 5 – Attributes: I determined the information to be kept for each entity. For instance, a band member needs to have a member id, name of the member, role of the member, email, and date of joining. Each entity was given a unique identification number to be used as a primary key.

Research Sources

Source 1 – Coldplay Official Website:

I reviewed Coldplay's official website and found that the site organizes information about albums and songs. This helped me determine that Album and Song should be entities in my Music Band Website database.

<https://www.coldplay.com/music/>

Source 2 – Coldplay Tour Website:

I reviewed Coldplay's official tour website and found that it provides information about concerts, including dates, venues, and cities. This helped me determine that Event and Venue should be entities in my database.

<https://www.coldplay.com/tour/>

Source 3 – Coldplay Official Website:

I reviewed Coldplay's official website and found information about the band and its members. This helped me determine that Band Member should be an entity in the database so information about each member can be stored.

<https://www.coldplay.com/>

Database Implementation

The database for the Band Music Website was created using MySQL Workbench software, which has six tables: Band Member, Album, Song, Venue, Event, and Member Event. In the process of creating the database, the keys, both primary and foreign, were applied to provide links among the tables and establish relationships. The database was created on a MySQL server in the process of forward engineering the EER diagram.

Sample Data and Testing

Sample data was introduced into the database to make sure the tables and relationships work correctly. The created database included information regarding band members, albums, songs, events, venues, and Member Event records. The testing employed SELECT and JOIN queries to ensure that data was stored properly and that relations between the tables worked as intended.

The obtained results confirmed that songs are linked to albums, events are linked to venues, and that there is a connection between members of the band and events through the Member Event table.

Conclusion

The project provided an example of utilizing a database to systematize information regarding a music band website. With MySQL Workbench, I was able to create the tables, relationships, and develop the database. Now this database is able to keep the records related to band members, albums, songs, places, and events. It also showed the way primary keys, foreign key and relationships are utilized to connect data in a relational database.